

Strategic Interaction among Public Innovation Hubs under Private Repricing

Abstract

Introduction. We study two regional governments that invest in public innovation intermediation while sharing a price-setting metropolitan private alternative. Cross-side participation can make public investments strategic complements when the private access price is held fixed, whereas endogenous private repricing adds an opposing strategic channel. **Methods.** We compare a fixed-price benchmark with the full sequential game, derive the public cross effects analytically around the zero-network benchmark, and use exact rational interval methods to certify the canonical interior equilibrium pair. **Results.** On regular interior equilibrium branches, the fixed-price public cross effect is zero at $\beta = 0$ but has a strictly positive first derivative in β , while the full-game cross effect is already strictly negative at $\beta = 0$. Hence there exists $\varepsilon > 0$ such that, for sufficiently small positive network interaction, the fixed-price best-response slope is positive while the full-game slope is negative. At the canonical parameter vector, an exact rational Krawczyk certificate and interval implicit differentiation verify the matched but distinct fixed-price and full-game equilibria, strict second-order conditions, and opposite best-response slopes. **Conclusion.** Endogenous commercial repricing can change the sign of decentralized public strategic interaction. The result is local rather than a global primitive-space classification, and welfare accounting is kept separate from the strategic reversal.

Keywords: public investment; two-sided platforms; strategic interaction; local public economics; endogenous pricing; innovation intermediation; network effects.

JEL Classification: H77; L13; D62; O38.

1 Introduction

Regional governments increasingly use public organizations to connect firms, universities, investors, experts, and other partners around innovation projects. At the same time, firms outside major cities can access dense commercial innovation ecosystems in metropolitan areas. These two forms of intermediation need not be pure competitors at the ecosystem level: they may collaborate, exchange participants, and provide complementary services. Yet for a focal project deciding which intermediary will be its primary route to partners, the alternatives can be substitutes. This coexistence raises a strategic question for decentralized public investment. When one region strengthens its own intermediation capacity, does that induce another region to invest more or less?

The answer is not immediate because innovation intermediation is two-sided. A regional investment that attracts or facilitates partners raises the value of the public route to projects. More projects then make the same route more attractive to additional partners. With a shared private alternative, an investment in one region also shifts projects and partners away from the private route. If the private access price is fixed, that weakening of the shared outside option can raise the marginal return to public investment in the other region. The public investments can therefore be strategic complements.

The private alternative, however, need not passively keep its price unchanged. A commercial innovation intermediary can respond to lost demand by cutting its access price. This response partly restores the private route’s attractiveness to projects in both regions. The induced repricing channel works against the positive cross-side network channel. The question of this paper is whether it can be strong enough to change the qualitative strategic relationship between the two decentralized public investors.

We develop a two-region model with two public innovation intermediaries and a shared metropolitan private intermediary. Regional governments simultaneously choose facilitation investments. The private intermediary then chooses a project-side access price. Projects select one primary intermediation route, while support-side partners may participate in multiple ecosystems. Project participation and partner participation reinforce one another. The governments maximize regional welfare; the private intermediary maximizes access-fee profit.

Our main result is a strategic sign reversal. We compare the full game with a nested benchmark that holds the private price fixed at the full-game equilibrium fee. On regular interior equilibrium branches around the zero-network benchmark, the two regimes point in opposite directions. In the fixed-price benchmark B3, the public cross derivative is exactly zero at $\beta = 0$, but its first derivative with respect to β is strictly positive. In the full game G, by contrast, the public cross derivative is already strictly negative at a symmetric regular state when $\beta = 0$. Continuity and strict second-order conditions therefore imply that there exists $\varepsilon > 0$ such that, for every sufficiently small positive β on the continuing regular branches,

$$\text{BR}_i^{B3'} > 0 > \text{BR}_i^{G'}.$$

Thus the reversal is an analytic local result, not a pattern inferred from a single numerical solution. The theorem is deliberately local: it does not characterize the entire primitive parameter region or provide an explicit formula for ε .

The economic reason for the reversal is transparent. With the private price fixed, cross-side participation transmits one region’s public expansion through the shared outside option and raises the other region’s marginal return to facilitation. Once the commercial intermediary is allowed to optimize its price, public expansion also changes private demand and therefore the private pricing policy. The resulting price response feeds back into project routing in both regions. In derivative form, the full-game public cross effect equals the fixed-price effect plus terms generated by the private price policy and its first and second derivatives. When those repricing terms dominate the fixed-price network force, the public game changes from strategic complementarity to strategic substitution. The negative follower-price contribution is not itself claimed as novel; the contribution is the qualitative reversal in the same public–public strategic object when the private pricing policy is activated.

We complement the analytic theorem with an exact certificate at the canonical interior parameter vector. The finite-decimal primitives are represented as exact rationals. A coupled rational Krawczyk argument isolates one G equilibrium and one distinct B3 equilibrium while imposing the same algebraic full-game price in the matched benchmark. Exact interval implicit differentiation then certifies the relevant public own and cross derivatives, the private price second-order condition, the negative price response, and local nonsingularity. Hence the reported sign reversal is not a floating-point, finite-difference, solver-tolerance, or root-selection artifact. This computer-assisted proposition is intentionally narrower than the analytic theorem: it certifies one economically relevant algebraic equilibrium pair and is not a primitive-space classification.

The earlier numerical exercises retain a different role. They display magnitudes and confirm that the certified equilibrium is not a knife-edge numerical presentation. The canonical parameter vector

is a constructive benchmark, not an empirical calibration, and small deterministic perturbations preserve the sign pattern. These computations supplement rather than prove the analytic result. We do not claim a global sign theorem, arbitrary-distribution robustness, or arbitrary nonlinear-network robustness.

Welfare is deliberately separated from strategic interaction. The private access fee is a regional project outflow but an aggregate transfer to the private intermediary, so changing the fee does not mechanically change national welfare. At the reported full-game configuration, a marginal increase in one region’s public investment creates a positive effect on the other region’s reduced welfare that is only partly offset by lower private profit. The resulting national coordination wedge is positive. This does not mean that strategic complements are socially good or strategic substitutes socially bad, and the paper does not establish a welfare-sign reversal.

The institutional interpretation is also intentionally narrow. Public innovation hubs and commercial metropolitan innovation communities in Japan and abroad exhibit the relevant ingredients: facilitation and matching, cross-side communities, paid access for some private organizations, and access by projects outside the metropolitan core. These examples support the plausibility of the primitives but do not establish the model’s causal repricing prediction. In particular, project single-homing is a focal-project primary-route abstraction, and private profit maximization is a reduced-form representation of a selected commercial operator class.

Our contribution is not the discovery of network complementarity, endogenous follower pricing, mixed public/private competition, or two-sided platforms. Each of those components has close precedents. Sequential-investment work such as [López and Vives \[2019\]](#) already shows how an endogenous follower response enters an earlier investment incentive. Mixed public/private investment and regional infrastructure competition are studied by [Zogheib and Bourreau \[2021\]](#), [Mun \[2019\]](#), [Mun and Nakagawa \[2010\]](#), while two-sided platform pricing and investment interactions appear in [Farhi and Hagiü \[2008\]](#), [Liu et al. \[2026\]](#), [Hahn and Kim \[2026\]](#), [Bontems et al. \[2025\]](#), [He et al. \[2026\]](#). The narrower contribution is the nested benchmark result: the same public–public strategic relationship is positive when the shared private price policy is switched off at the matched full-game fee and negative when that price setter is activated. The analytic neighborhood result and exact canonical certificate strengthen the status of that comparison without converting the familiar component mechanisms into novelty claims.

The rest of the paper proceeds as follows. Section 2 presents the model. Section 3 characterizes participation, pricing, and the nested benchmarks. Section 4 develops the mechanism decomposition, states the analytic sign-reversal theorem, and reports the exact canonical certificate. Section 5 develops welfare and coordination. Section 6 separates analytic local persistence from numerical and functional-form robustness. Section 7 discusses institutional interpretation and empirical implications. Section 8 positions the result in the related literature. Section 9 discusses limitations and scope, and Section 10 concludes.

2 Model

We study two regions, indexed by $i \in \{1, 2\}$. Each regional government operates a public innovation intermediary H_i and chooses a facilitation investment $x_i \in [0, 1]$. The two regions also share access to a private metropolitan innovation intermediary H_T . The private intermediary charges a project-side access price $p_T \geq 0$. The institutional language is intended to capture settings in which local public organizations provide matching, coordination, events, and partner recruitment while projects may instead use a dense commercial innovation ecosystem in a major metropolitan area. The economic structure, rather than the institutional labels, is the object of the model.

2.1 Projects and intermediation routes

Each region contains a unit mass of projects with types $z \sim U[0, 1]$. A focal project selects one primary intermediation route from

$$\{0, H_1, H_2, H_T\},$$

where 0 is non-participation. This single-homing assumption applies to the primary route of a focal project, not to all activities of the firm that owns the project.

Let n_h^P denote the mass of support-side partners associated with route h . Define the project-side quality index

$$q_h \equiv v + \alpha n_h^P, \quad \alpha > 0.$$

A type- z project in region r obtains

$$U_{rh}(z) = zq_h - \kappa_{rh} - \mathbf{1}\{h = T\}p_T, \quad (1)$$

while $U_{r0} = 0$. Local public access has cost $\kappa_{rr} = \kappa_L$, access to the other regional public hub has cost $\kappa_{rj} = \kappa_L + \tau$ for $j \neq r$, and metropolitan private access has baseline cost $\kappa_{rT} = \kappa_T$. The parameter $\tau \geq 0$ makes the remote regional public hub weakly less attractive than the local public hub. In the central regime used for the main result, a region's projects sort among non-participation, the shared private route, and their local public route.

2.2 Support-side participation and public facilitation

Support-side partners may participate in multiple ecosystems. In the interior participation regime their masses satisfy

$$n_i^P = \rho + x_i + \beta n_i^F, \quad i = 1, 2, \quad (2)$$

$$n_T^P = \rho_T + \beta n_T^F, \quad (3)$$

where n_h^F is the mass of focal projects using route h . The constants ρ and ρ_T capture baseline partner attractiveness. The parameter $\beta \geq 0$ is the project-to-partner cross-side feedback. Thus an increase in project participation raises support-side participation, which in turn raises project value through αn_h^P .

Government i bears the convex facilitation cost

$$C_i(x_i) = \frac{\gamma}{2} x_i^2, \quad \gamma > 0. \quad (4)$$

The private intermediary has zero real operating cost in the baseline and earns

$$\Pi_T = p_T n_T^F. \quad (5)$$

The zero-cost assumption is a normalization for the reduced-form pricing problem, not a claim about the cost structure of actual innovation hubs.

2.3 Timing and equilibrium

The game has four steps. First, the two regional governments choose (x_1, x_2) . Second, after observing those investments, the private intermediary chooses p_T . Third, projects sort across routes and support-side participation satisfies the cross-side fixed point. Fourth, project surplus, partner surplus, regional welfare, and private profit are realized. We study subgame-perfect equilibrium.

The main result is evaluated in a smooth central allocation regime. Write $a = \kappa_T + p_T$. For a resident project of region i , the relevant ordering is

$$0 \longrightarrow H_T \longrightarrow H_i,$$

with cutoffs

$$s = \frac{a}{q_T}, \quad t_i = \frac{\kappa_L - a}{q_i - q_T}, \quad (6)$$

satisfying $0 < s < t_i < 1$, $q_i > q_T$, and $a < \kappa_L$. Hence projects with $z < s$ remain outside, projects with $s < z < t_i$ use H_T , and projects with $t_i < z < 1$ use H_i . These inequalities are strict at the numerical configuration used below.

3 Equilibrium and Benchmarks

This section reduces the participation subgame to a smooth system and defines the two nested environments used to identify the role of private repricing.

3.1 Participation fixed point

Let

$$\delta = \alpha\beta, \quad A_i = v + \alpha(\rho + x_i), \quad A_T = v + \alpha\rho_T, \quad d = \kappa_L - \kappa_T - p_T.$$

In the central regime, local project mass is $n_i^F = 1 - t_i$ and private project mass is

$$n_T^F = t_1 + t_2 - \frac{2(\kappa_T + p_T)}{q_T}.$$

Using (2)–(3), the project-quality indices satisfy

$$q_i = A_i + \delta(1 - t_i), \quad (7)$$

$$q_T = A_T + \delta \left(t_1 + t_2 - \frac{2(\kappa_T + p_T)}{q_T} \right). \quad (8)$$

Together with the indifference condition $t_i(q_i - q_T) = d$, the participation equilibrium solves

$$F_i \equiv t_i [A_i + \delta(1 - t_i) - q_T] - d = 0, \quad i = 1, 2, \quad (9)$$

$$F_T \equiv q_T - A_T - \delta \left[t_1 + t_2 - \frac{2(\kappa_T + p_T)}{q_T} \right] = 0. \quad (10)$$

This three-equation system is the smooth object used for implicit differentiation and numerical verification.

3.2 Private pricing

Let $D_T \equiv n_T^F$ denote private-route project demand after participation has adjusted. An interior private price satisfies

$$D_T + p_T D_{T,p} = 0, \quad (11)$$

with second-order condition

$$2D_{T,p} + p_T D_{T,pp} < 0. \quad (12)$$

Implicit differentiation of (11) yields the policy response

$$p_{T,x_i}^* = -\frac{D_{T,x_i} + p_T D_{T,px_i}}{2D_{T,p} + p_T D_{T,pp}}. \quad (13)$$

At the verified full-game configuration this derivative is negative. Economically, stronger regional facilitation draws projects away from the shared private route; the private intermediary responds by lowering its optimal access price.

The negative price response is not itself a new mechanism. Indeed, when $\beta = 0$ the private demand problem becomes linear and the sign of the price response can be established directly. The role of two-sidedness in our main result is different: it creates the positive public–public cross effect in the fixed-price benchmark.

3.3 Nested benchmarks

We compare two environments with identical primitives.

Fixed-price benchmark (B3). Both regional public hubs and the shared private hub operate, and participation remains two-sided, but the private access price is fixed at $\bar{p}_T$. For the headline comparison we set

$$\bar{p}_T = p_T^G,$$

the equilibrium private price in the full game. This is a matched-price local diagnostic. At the certified comparison, it holds the fee level at the value chosen in G while suppressing the policy response $p_T^*(x_1, x_2)$. The benchmark therefore separates the local strategic contribution of repricing from a mechanical difference in the fee level. It does not assert that the two public investments are complements for every exogenous value of $\bar{p}_T$, and it is not interpreted as a price-regulation counterfactual.

Full game (G). The private intermediary observes (x_1, x_2) and optimally chooses p_T according to (11). Governments anticipate the reduced price policy $p_T^*(x_1, x_2)$ when choosing facilitation investments.

Let $\widetilde{W}_i$ denote government i 's reduced objective after downstream private pricing and participation have been solved. At an interior local optimum,

$$\text{BR}'_i(x_j) = -\frac{\widetilde{W}_{i,x_i x_j}}{\widetilde{W}_{i,x_i x_i}}. \quad (14)$$

Because the own second derivative is negative at the regular interior configurations considered below, the sign of the best-response slope is the sign of the public cross derivative.

4 Strategic Interaction and Private Repricing

This section separates three objects that play different roles in the argument: the fixed-price network force, the additional repricing force, and the resulting sign comparison between B3 and G. The first two explain the mechanism; the third is the main theorem.

4.1 Fixed private price: cross-side reinforcement

Let

$$M_{B3} \equiv W_{i,x_i x_j}^{B3}$$

denote the public cross derivative after participation has been solved while the private access price is held fixed. Under a strict public second-order condition, (14) implies that M_{B3} and the B3 best-response slope have the same sign.

When p_T is fixed, an increase in x_j raises support-side participation at H_j , attracts projects toward the regional public route, and reduces project and partner participation at the shared private route. The weakened shared outside option then raises the marginal value of public facilitation in region i . Cross-side participation is the only cross-region link in this benchmark central regime.

Proposition 1 (First-order benchmark complementarity). *Consider a regular interior B3 branch through $\beta = 0$ with $d = \kappa_L - \kappa_T - \bar{p}_T > 0$ and beta-zero quality gaps $\Delta_i = A_i - A_T > 0$. Then*

$$M_{B3}(0) = 0$$

and

$$\left. \frac{\partial M_{B3}}{\partial \beta} \right|_{\beta=0} = \frac{\alpha^3 d^3}{\Delta_i^3 \Delta_j^2} > 0. \quad (15)$$

Consequently, if the branch and its strict public second-order condition continue smoothly, public investments are strategic complements for all sufficiently small positive β .

The result identifies exactly what two-sidedness contributes. At $\beta = 0$, B3 has no public–public cross effect. A positive network interaction turns that zero cross effect positive to first order. Because $M_{B3}(0, x, p)$ is identically zero within the beta-zero central regime, derivatives of the B3 equilibrium path and of the matched price do not enter the first derivative in (15); the positive term is the direct first-order cross-side effect.

4.2 Endogenous private price: the repricing channel

In the full game, government i evaluates

$$\widetilde{W}_i(x_i, x_j) = W_i(x_i, x_j, p_T^*(x_i, x_j)).$$

Write $p_i = p_{T,x_i}^*$, $p_j = p_{T,x_j}^*$, and $p_{ij} = p_{T,x_i x_j}^*$. Exact differentiation gives the decomposition

$$M_G = M_{\text{fixed}} + P_{\text{price}}, \quad P_{\text{price}} = W_{i,x_i p} p_j + W_{i,x_j p} p_i + W_{i,pp} p_i p_j + W_{i,p} p_{ij}, \quad (16)$$

where $M_{\text{fixed}} = W_{i,x_i x_j}$ is the cross derivative that would remain if the price policy were locally switched off at the same state.

Proposition 2 (Private-price feedback). *At any regular full-game state with $M_{\text{fixed}} > 0$, endogenous repricing reverses the sign of the public cross derivative if and only if*

$$-P_{\text{price}} > M_{\text{fixed}} > 0.$$

Thus repricing changes strategic complementarity into strategic substitution precisely when the price-policy contribution dominates the positive fixed-price force at that state.

Proposition 2 is a mechanism identity rather than a novelty claim about follower pricing. Economically, the chain is

$$\begin{aligned} x_j \uparrow &\implies \text{private demand falls} \\ &\implies p_T^* \text{ adjusts} \\ &\implies \text{project routing in region } i \text{ changes} \\ &\implies W_{i,x_i} \text{ changes.} \end{aligned}$$

The production model predicts $p_i < 0$ at the canonical full-game equilibrium: stronger regional facilitation induces the private intermediary to cut its price. This price cut restores part of the private route’s attractiveness and works against the positive fixed-price network force.

4.3 Analytic strategic reversal

The beta-zero limit makes the contrast between the two forces especially sharp. In B3 the cross effect starts from zero and moves positively with network interaction. In G, private repricing already couples the governments when $\beta = 0$.

Theorem 1 (Local strategic sign reversal). *Suppose there exist regular interior B3 and G equilibrium branches at $\beta = 0$. Assume the relevant participation and equilibrium Jacobians are nonsingular, the private optimum and both public optima satisfy strict second-order conditions, and the branches continue smoothly for small β . Let the B3 benchmark use the matched full-game price. Then there exists $\varepsilon > 0$ such that, for every $\beta \in (0, \varepsilon)$ on these continuing branches,*

$$\text{BR}_i^{B3'} > 0 > \text{BR}_i^{G'}.$$

More specifically, B3 satisfies (15), while at a symmetric regular beta-zero full-game state, with $T = A_T = v + \alpha\rho_T > 0$ and $\Delta = A_i - A_T > 0$,

$$M_G(0) = -\frac{3T\alpha^2\kappa_L^2}{16\Delta(\Delta + T)^3} < 0. \quad (17)$$

The proof is in Appendix A.3. The substantive point is the asymmetric beta-zero limit. B3 requires cross-side interaction to create a positive link, whereas G has a strictly negative public cross effect even before that link is activated. Strict signs, strict second-order conditions, and regularity then preserve the ordering for sufficiently small positive β . The theorem is sufficient and local. It does not provide a primitive formula for ε or a necessary-and-sufficient characterization of the entire parameter space.

Corollary 1 (Strategic effect of commercial adaptation). *Holding the private alternative fixed can give the wrong qualitative description of decentralized public competition. Activating the private intermediary’s optimizing price policy can change not only the magnitude but also the sign of the public–public strategic relationship.*

4.4 Exact certification at the canonical equilibrium

The analytic theorem establishes a nonempty local mechanism class subject to regular beta-zero branches. A separate computer-assisted result verifies that the economically relevant canonical comparison used in the paper satisfies the full equilibrium, matching, curvature, and sign conditions exactly rather than only to floating-point tolerance.

Table 1: Strategic interaction under fixed and endogenous private pricing

Regime	x	BR slope
G	0.684028	-0.024120
B3	0.656020	0.114436

Proposition 3 (Exact certification at the canonical parameter vector). *For the canonical finite-decimal primitive vector, interpreted as exact rational numbers, there is a unique coupled G/B3 stationary solution inside the rational isolating box reported in Appendix B. The B3 and G public equilibria are distinct and share the same algebraic full-game price in the matched benchmark. Exact rational interval implicit differentiation certifies*

$$H_{11}^{B3} < 0 < H_{12}^{B3}, \quad H_{11}^G < 0, \quad H_{12}^G < 0,$$

as well as a strict private price second-order condition, $p_{T,x_i}^* < 0$, and the relevant nonsingularity conditions. Hence

$$\text{BR}_i^{B3'} > 0 > \text{BR}_i^{G'}$$

holds at that algebraic equilibrium pair.

Proposition 3 is an exact sign-at-algebraic-root statement, not a general primitive theorem. Its purpose is to rule out solver error, finite-difference error, tolerance dependence, and inconsistent root selection. The high-degree elimination objects needed internally for root isolation are not part of the economic contribution and are left in the machine-readable repository artifacts.

Table 1 reports the corresponding magnitudes from the generated production pipeline. The parameter vector is a constructive benchmark, not an empirical calibration.

The fixed-price best-response slope is approximately 0.1144, while the full-game slope is approximately -0.0241 . The matched private fee is the same object in the two comparisons, but the public equilibrium investments are correctly re-solved and differ across B3 and G. The private-price response is approximately -0.0165 . These magnitudes illustrate the mechanism; Theorem 1 supplies the analytic local result and Proposition 3 supplies the exact canonical verification.

5 Welfare and Coordination

The sign of a government's best-response slope is not a welfare ranking. Strategic interaction describes how one government's optimal investment responds to the other's choice; welfare depends on project allocation, partner surplus, real facilitation costs, and private profit. We therefore keep the strategic reversal and the welfare analysis separate.

5.1 Regional and aggregate welfare

In the central regime, project surplus in region i is

$$PS_i = \int_s^{t_i} (zq_T - \kappa_T - p_T) dz + \int_{t_i}^1 (zq_i - \kappa_L) dz. \quad (18)$$

With uniform support-side costs, the total surplus associated with partner participation at route h is $(n_h^P)^2/2$. Under the symmetric regional attribution used in the regional objectives, each region

Table 2: Welfare at the numerical witness			
Regime	W_i	Π_T	W^N
G	0.223019	0.012603	0.458642
B3	0.210911	0.013891	0.435714

receives one half of that national partner surplus. Regional welfare is therefore

$$W_i = PS_i + \frac{(n_1^P)^2 + (n_2^P)^2 + (n_T^P)^2}{4} - \frac{\gamma}{2}x_i^2. \quad (19)$$

Aggregate welfare is

$$W^N = W_1 + W_2 + \Pi_T. \quad (20)$$

A project using H_T pays p_T in regional project surplus, while the same payment appears as private profit. Because private operating cost is normalized to zero, the transfer cancels exactly at the aggregate level:

$$-p_T n_T^F + \Pi_T = 0.$$

Accordingly, a lower private price does not mechanically imply higher aggregate welfare.

5.2 A cross-regional coordination wedge

At an interior decentralized full-game equilibrium, government i 's own reduced first-order condition is zero. Hence the national marginal effect of a coordinated increase in x_i can be written as

$$\frac{dW^N}{dx_i} = \frac{d\widetilde{W}_j}{dx_i} + \frac{d\Pi_T}{dx_i}, \quad j \neq i. \quad (21)$$

At the verified configuration the rival-region welfare effect is approximately 0.44471, while the private-profit effect is approximately -0.02194 . Their sum is

$$\frac{dW^N}{dx_i} \simeq 0.42277 > 0.$$

Thus a real cross-regional coordination externality remains after the private-fee transfer has been removed. Its sign is not implied by the sign of BR'_i .

5.3 Controlled B3–G comparison

Table 2 reports the generated local welfare comparison. B3 fixes the private price at the equilibrium fee of G; it is an identification benchmark for price endogeneity, not a literal price-regulation experiment.

At this configuration aggregate welfare is higher in G than in B3, but this is a local numerical comparison, not a theorem of global welfare dominance. In particular, Theorem 1 does not imply that strategic substitutability is socially desirable, nor does it establish a reversal in the sign of the welfare externality.

6 Robustness

The robustness analysis has an expository role distinct from the proof. Theorem 1 already establishes that the sign reversal holds on a nonempty interval of sufficiently small positive network interaction, conditional on regular beta-zero branches and strict second-order conditions. Proposition 3 separately establishes the reversal exactly at the canonical interior vector. The computations below ask whether the displayed economic configuration is numerically knife-edge and how far the maintained local interpretation can be extended without changing the model.

Network interaction. The analytic result organizes variation in β near zero: B3 starts from a zero cross effect with a strictly positive β derivative, while G starts from a strictly negative cross effect. The canonical benchmark uses $\beta = 0.05$ and satisfies the reversal by exact certification. Because the theorem leaves ε implicit, we do not infer that every value between zero and 0.05 lies on one globally regular branch solely from these two facts. That stronger connected-interval statement would require additional branch tracking and is not needed for the present local theorem.

Primitive perturbations. All inequalities defining the canonical configuration are strict. The project cutoffs and partner masses are interior, the relevant public and private second-order conditions are negative, and both best-response slopes are bounded away from zero. The participation system and first-order conditions are smooth with nonsingular local Jacobians. A deterministic perturbation exercise that varies the primitive vector by ± 0.5 percent in 20 reproducible draws preserves

$$\text{BR}_i^{B3'} > 0 > \text{BR}_i^{G'}$$

in all 20 cases. This exercise is not the source of the theorem; it confirms that the reported benchmark magnitudes are not a tolerance-level accident.

Distribution and network functional form. The uniform project-type distribution is not locally essential. Replacing it by a sufficiently small C^1 perturbation with positive density at the relevant cutoffs changes project masses and surplus terms smoothly. Conditional on the same interior ordering and nonsingularity, the strict sign inequalities persist by the implicit function theorem. The linear partner-participation specification is likewise locally nonessential: small C^1 perturbations of the network functions that preserve the relevant derivative signs and regularity induce continuous changes in the same equilibrium system.

Asymmetry and scope. Small asymmetric perturbations of the action state around the symmetric canonical configuration preserve the strict sign reversal by continuity. This is not a theorem for arbitrary region-specific primitives. The robustness claims remain deliberately local: we do not establish a global parameter region, arbitrary-distribution result, arbitrary nonlinear-network result, or fully heterogeneous-regions theorem. Broad parameter-plane classification would add substantial algebra and computation without strengthening the paper’s core mechanism proportionately.

7 Institutional Interpretation and Empirical Implications

The model is motivated by an institutional class in which regional public organizations perform innovation intermediation while projects can also use priced commercial innovation ecosystems in

large metropolitan areas. The examples in this section establish the plausibility of the primitives; they do not provide causal evidence for the model’s comparative statics.

7.1 Public and private innovation intermediation

In Japan, CIC Tokyo describes a paid coworking and innovation community that connects startups with investors, government, professional service providers, and other participants; its official materials also describe use by people who normally work in local areas but need a Tokyo base.¹ SHIBUYA QWS is a membership-based co-creation facility with project and corporate membership categories, and its membership agreement expressly permits changes to membership plans and charges under specified operational or economic conditions.² These examples make a priced metropolitan outside option and operator discretion over an effective access price institutionally plausible. They do not establish that the observed price is chosen by the exact profit function in (5).

The same institutional class is not uniquely Japanese. Level39 in London offers paid community and coworking memberships and describes connections to mentors, investors, and industry experts.³ CIC locations in Rotterdam, Warsaw, Cambridge, and Boston similarly combine paid workspace access with innovation-community and partner-network functions. On the public side, Helsinki’s Campus Incubators Programme describes a city role in funding and connecting university incubation activity to the wider innovation ecosystem, while Barcelona Activa operates municipal startup incubation and network-support programs.⁴

These institutions need not be ecosystem-wide competitors. Public and private organizations may collaborate, share events, or provide complementary services. The model requires a narrower form of substitution: for a focal project choosing a primary intermediation route, improving one route may reduce use of another. This distinction allows ecosystem-level complementarity to coexist with project-level substitution.

7.2 Homing and objective assumptions

Project single-homing should be interpreted as a primary-route abstraction. A firm may participate in many networks, rent multiple workspaces, or maintain relationships with several intermediaries while a particular collaboration or commercialization project relies on one main route. The support side is treated differently: investors, experts, universities, professional service organizations, and public agencies can plausibly connect to more than one innovation ecosystem. The model therefore permits partner multihoming, without claiming that every observed partner simultaneously participates in every relevant hub.

The private profit objective is also reduced form. It selects a commercial operator class for which paid access and demand responsiveness are economically meaningful; it does not assert that all real-world innovation hubs maximize the one-period expression $p_T n_T^F$ or lack ecosystem objectives.

¹CIC Tokyo, [Coworking](#) and [launch/company description](#), official webpages. Accessed September 3, 2026.

²SHIBUYA QWS, [Membership](#) and [membership agreement](#), official webpages. Accessed September 3, 2026.

³Level39, [Community Membership](#) and [About](#), official webpages. Accessed September 3, 2026.

⁴City of Helsinki, [Campus Incubators Programme](#); Barcelona Activa, [Startup Lab](#). Official webpages, accessed September 3, 2026.

7.3 Model predictions

The model’s primary price prediction is

$$p_{T,x_i}^* < 0$$

at the verified configuration. Stronger regional facilitation lowers demand for the private route, and the optimizing private intermediary responds by cutting its access price. This is a model prediction, not an observed historical fact about CIC Tokyo, QWS, Level39, or any other named institution.

A second implication concerns strategic responses across governments. When private pricing is relatively rigid, cross-side participation can make regional investments move in the same direction. Greater private pricing flexibility introduces a countervailing reallocation channel and can, under the conditions identified in Theorem 1, reverse that response. Empirically distinguishing the two regimes would require data on regional facilitation, project routing, partner participation, and private access prices. Such a test lies beyond the present theory paper.

8 Related Literature

Our result combines ingredients that are individually familiar. The contribution is therefore best located by separating the known components from the qualitative comparison that remains.

Endogenous follower responses and investment incentives. The closest generic threat is López and Vives [2019]. Their sequential-investment framework makes clear that a downstream or follower response can enter an earlier investment incentive; the existence of an endogenous price-response term is therefore not new here. Our narrower object is the strategic relationship between *two* decentralized public investors. We hold the third party’s equilibrium price fixed in a nested benchmark and then restore its optimizing price policy. The same pair of public investments is complementary in the former environment and substitutable in the latter. The comparison thus asks whether activating an already-understood response channel changes the qualitative game played by two other strategic actors.

Public and private investment competition. Zogheib and Bourreau [2021] study public versus private investment in network industries, and Mun [2019] and Mun and Nakagawa [2010] analyze investment and pricing in transport settings with public or regional strategic objectives. These papers make clear that mixed objectives and decentralized regional investment are not themselves novel. Relative to this line, the present model emphasizes cross-side participation and a shared price-setting outside option whose response alters the sign of public–public strategic interaction.

Two-sided platforms and network interaction. Strategic interaction in two-sided market oligopoly is developed in Farhi and Hagiū [2008]. More recent work studies public and private payment platforms [Liu et al., 2026], platform investment under contractual pricing constraints [Hahn and Kim, 2026], sequential pricing on multisided platforms [Bontems et al., 2025], and pricing and service investment under mixed network externalities [He et al., 2026]. These contributions absorb important components of our environment: network feedback, platform pricing, and strategic investment. We do not claim novelty for those components.

Qualitative effects of sequential pricing. Sánchez-Cartas [2026] provides another nearby illustration of how follower responses can alter qualitative outcomes in sequential platform settings.

Again, the distinction in the present paper is the nested fixed-price versus optimizing-price comparison for the *public–public* best-response slope. Among the closest models examined here, we do not identify a result that supplies this whole matched-price public–public sign comparison by direct relabeling or restriction. This positioning is intentionally narrow: it does not imply a new general theory of public platforms or two-sided markets.

9 Discussion

The model isolates one mechanism: endogenous repricing by a shared private intermediary can change the strategic relation between decentralized public investments. Several boundaries matter for interpreting that result.

First, the analytic result is local rather than global. Theorem 1 establishes the sign reversal for sufficiently small positive β around regular beta-zero B3 and G branches with strict second-order conditions and nonsingular local systems. It does not provide a primitive formula for the endpoint ε , prove that such regular branches exist for every primitive vector, or characterize all primitives for which reversal occurs. The exact canonical certificate closes the equilibrium and sign obligations at one economically relevant rational vector, but it is likewise not a primitive-space classification. These limitations are compatible with the paper’s mechanism claim; they rule out interpreting the result as a global sign theorem.

Second, the novelty is combinational and therefore exposed to a natural objection. Positive cross-side network feedback is familiar, and negative effects transmitted through an endogenous follower price are familiar. The economic content of the paper is the nested comparison showing that these forces can change the *type of strategic game* played by two public investors. The analytic beta-zero limit sharpens that point: the fixed-price public cross effect needs positive network interaction to emerge, whereas the full game is already negatively coupled through commercial adaptation at the same limit. The claim is accordingly limited to that qualitative interaction; it does not relabel either familiar component as new.

Third, the computer-assisted certificate is evidence of proof credibility rather than mathematical novelty. Exact rational root isolation and interval implicit differentiation are used because the canonical stationary system is algebraic and has multiple potential roots. They establish that the reported equilibrium comparison, curvature conditions, and derivative signs do not depend on floating-point root selection or finite differences. The underlying high-degree elimination objects are intentionally kept out of the economic argument.

Fourth, primary-route single-homing is a modeling abstraction. Innovation projects and the firms behind them can interact with several networks. The maintained interpretation is narrower: a focal project has one principal intermediation route for the activity being modeled. Allowing project multihoming could attenuate route substitution and lies outside the present analysis.

Fifth, the private objective and instrument set are reduced-form approximations. Real commercial innovation communities may value occupancy, membership revenue, sponsorship, ecosystem growth, or longer-run strategic goals, and may use richer contracts than a single posted price. Similarly, a posted membership or access fee may bundle workspace and community services. The model uses a one-dimensional effective access price to represent the margin through which a commercial alternative responds to project demand; it does not claim robustness to arbitrary contracting environments. The sequence in which public facilitation is chosen before private pricing is likewise a maintained feature of the research question rather than a claim about all institutional timings.

Sixth, the institutional evidence establishes an analogue class, not an observed treatment-control relationship. We do not observe a causal episode in which expansion of a regional public hub induced

CIC Tokyo, QWS, Level39, or another named operator to cut its price. The negative price response is a theoretical prediction. A useful empirical design would combine changes in regional facilitation capacity with project routing, partner participation, and flexible private access prices.

Finally, strategic substitution is not a policy diagnosis. The fixed-price environment is an identification benchmark, not a recommendation to regulate private prices. Nor does the positive coordination wedge at the reported configuration imply that any specific merger, subsidy, mandate, or price restriction is optimal. The model’s policy content is limited to identifying a coordination externality and a private repricing channel that a richer policy analysis would need to take into account.

10 Conclusion

This paper studies two decentralized regional governments that invest in public innovation intermediation while sharing a price-setting metropolitan private alternative. Cross-side participation links project routing to partner participation. When the private access price is held fixed, this link can make the regional public investments strategic complements. When the private intermediary optimally reprices, its price response can dominate that positive network channel and turn the same investments into strategic substitutes.

The main sign reversal is not merely numerical. Around regular beta-zero equilibrium branches, the fixed-price public cross effect starts from zero and increases strictly with network interaction, while the full-game cross effect is already strictly negative. This yields an analytic local theorem: for sufficiently small positive β , the B3 and G best responses have opposite signs. At the canonical rational parameter vector, a separate exact computer-assisted certificate jointly isolates the distinct matched B3 and G equilibria and verifies the strict curvature and derivative signs. The paper therefore closes the main theoretical claim at two complementary levels without claiming a global primitive-space characterization.

Welfare accounting further shows that the strategic sign is distinct from the welfare externality: the private access fee is an aggregate transfer, while decentralized public investment can generate a real cross-regional coordination wedge. Institutional examples make the underlying class of public and commercial innovation intermediaries plausible, while the price response itself remains a testable model prediction.

The broader lesson is limited but useful. When decentralized public intermediaries share an adaptive private outside option, treating that outside option as fixed can give the wrong qualitative picture of public strategic interaction. The economic issue is not the presence of one more endogenous variable; it is the pricing feedback through which the shared private alternative adapts to public investment. How often this mechanism is quantitatively important is an empirical question.

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Data Availability Statement

No external datasets were generated or analyzed in this theoretical study. The symbolic, numerical, and computer-assisted verification code underlying the reported results is included in the reproducibility materials accompanying the manuscript.

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A Derivations and Proofs

A.1 Project cutoffs

For route h , write $a_{rh} = \kappa_{rh} + \mathbf{1}\{h = T\}p_T$. The outside cutoff is

$$z_{h0}^r = \frac{a_{rh}}{q_h},$$

and the pairwise cutoff between routes h and g , whenever $q_h \neq q_g$, is

$$z_{hg}^r = \frac{a_{rh} - a_{rg}}{q_h - q_g}.$$

Under the central ordering used in the main text, these expressions reduce to the cutoffs in (6).

A.2 Private-price response

Differentiating the private first-order condition (11) with respect to x_i gives (13). When $\beta = 0$, private demand is linear. Let Δ_i denote the corresponding local-private quality difference and define

$$A = \frac{1}{\Delta_1} + \frac{1}{\Delta_2}, \quad B = \frac{2}{q_T}.$$

The interior private price is

$$p_T^* = \frac{(\kappa_L - \kappa_T)A - \kappa_T B}{2(A + B)}.$$

Then

$$\frac{\partial p_T^*}{\partial A} = \frac{\kappa_L B}{2(A + B)^2} > 0, \quad A_{x_i} = -\frac{\alpha}{\Delta_i^2} < 0,$$

so $p_{T,x_i}^* < 0$. This limiting case shows that endogenous private pricing does not require two-sided participation. Two-sidedness is instead essential to the positive B3 cross effect used in Theorem 1.

A.3 Proof of the local strategic sign reversal

Let $\delta = \alpha\beta$, $T = A_T = v + \alpha\rho_T$, $\Delta_i = A_i - A_T$, and $d = \kappa_L - \kappa_T - p_T$. On a regular interior beta-zero branch, $T > 0$, $\Delta_i > 0$, and $d > 0$. Write $M_{B3} = W_{i,x_i x_j}^{B3}$ and let M_G denote the corresponding reduced full-game cross derivative.

At $\delta = 0$, the B3 participation equations separate across regions once the shared private price is held fixed. Hence

$$M_{B3}(0, x, p) = 0$$

throughout the relevant beta-zero central regime. A first-order expansion of the exact participation system and the regional objective gives

$$\left. \frac{\partial M_{B3}}{\partial \delta} \right|_{\delta=0} = \frac{\alpha^2 d^3}{\Delta_i^3 \Delta_j^2} > 0.$$

Since $\delta = \alpha\beta$,

$$\left. \frac{\partial M_{B3}}{\partial \beta} \right|_{\beta=0} = \frac{\alpha^3 d^3}{\Delta_i^3 \Delta_j^2} > 0.$$

Because $M_{B3}(0, x, p)$ is identically zero, differentiating along a continuing B3 equilibrium and matched-price path introduces no additional first-order terms at zero: the chain-rule multipliers on the path derivatives are zero.

For G, the private price policy remains an active strategic link at $\beta = 0$. At a symmetric regular beta-zero state let $\Delta = A_i - A_T > 0$. Substituting the exact interior private optimum into the full chain-rule decomposition (16) yields

$$M_G(0) = -\frac{3T\alpha^2\kappa_L^2}{16\Delta(\Delta + T)^3} < 0.$$

Thus B3 has a zero cross effect with a strictly positive derivative in β , whereas G has a strictly negative cross effect at the limiting state itself.

By the maintained nonsingularity conditions, the implicit function theorem supplies locally smooth B3 and G equilibrium branches and smooth reduced derivatives. Strict public and private second-order conditions and the central-regime inequalities also persist locally. Therefore there is an $\varepsilon > 0$ such that, for every $\beta \in (0, \varepsilon)$ on the continuing branches,

$$M_{B3}(\beta) > 0 > M_G(\beta).$$

Under the strict public second-order conditions, the best-response slope in (14) has the sign of the relevant public cross derivative. Hence

$$\text{BR}_i^{B3'} > 0 > \text{BR}_i^{G'},$$

which proves Theorem 1. The argument is sufficient and local. It does not produce a primitive formula for ε , nor does it establish existence and strict second-order conditions for every primitive vector.

B Exact Canonical Certificate

The canonical finite-decimal primitives are interpreted as exact rational numbers. The certification proceeds on a symmetry-reduced algebraic representation of the participation and stationary conditions. Rather than solving a high-degree root by radicals, an exact rational Krawczyk operator is applied to the coupled G/B3 system. It proves that there is one and only one solution in the box

$$\begin{aligned} 0.5848900730 &< t_G < 0.5848900731, \\ 0.1388515735 &< q_G < 0.1388515736, \\ 0.0227466514 &< p_G < 0.0227466515, \\ 0.6100908625 &< t_B < 0.6100908626, \\ 0.1402669351 &< q_B < 0.1402669352. \end{aligned}$$

Reconstruction gives $x_G \simeq 0.684028196361275$ and $x_B \simeq 0.656020390747393$. Thus the certificate imposes one common algebraic p_G while correctly allowing the public equilibrium investment to differ across G and the matched-price B3 benchmark.

Exact rational interval implicit differentiation on this isolating box gives

$$\begin{aligned} -0.118357218885676 &< H_{11}^{B3} < -0.118357218885651, \\ 0.0135431333332136 &< H_{12}^{B3} < 0.0135431333332153, \\ -0.199391062101166 &< H_{11}^G < -0.199391062101008, \\ -0.00480853615115529 &< H_{12}^G < -0.00480853615112228. \end{aligned}$$

The private-price second derivative lies in

$$(-52.1627947003595, -52.1627947003591),$$

and the public-investment response of the private price satisfies

$$p_{T,x_i}^* \in (-0.0165046011641228, -0.0165046011641190).$$

The resulting best-response slopes satisfy

$$\begin{aligned} \text{BR}_i^{B3'} &\in (0.114425917242068, 0.114425917242105), \\ \text{BR}_i^{G'} &\in (-0.024116106812849, -0.024116106812664). \end{aligned}$$

The same interval calculation verifies the required local Jacobian nonsingularity conditions. These strict rational intervals establish Proposition 3 without finite differences or floating-point sign decisions.

The coupled certification is reproduced by `analytic.level3_reopen/code/verify_l31.py`. Internally, algebraic elimination identifies a high-degree component before root isolation, but the coefficients of that polynomial are not economically informative and are not reproduced here. The certificate is deliberately a sign-at-algebraic-root result for one exact rational primitive vector. It is not a primitive-parameter cell decomposition or a necessary-and-sufficient global characterization.

C Numerical Verification and Local Robustness

The production numerical pipeline remains useful for magnitudes and independent consistency checks. It solves the participation equations (9)–(10), the private pricing condition (11), and the two public first-order conditions at the benchmark vector. The numerical gate requires the participation and decision first-order conditions to satisfy documented tolerances; the central allocation cutoffs and project and partner masses to be interior; private and public second-order conditions to be negative; local participation and equilibrium Jacobians to be nonsingular; and the B3 and G public cross derivatives to have strict opposite signs.

The production numerical values are close to the exact certified intervals above. Their role is expository rather than foundational: Theorem 1 supplies the analytic local mechanism, and Proposition 3 supplies the exact canonical sign verification. Floating-point comparisons use explicit tolerances rather than string equality.

A deterministic 20-draw perturbation audit moves the primitive vector by up to ± 0.5 percent and verifies that all draws retain the headline sign pattern. This is computational confirmation that the displayed benchmark is not numerically knife-edge; it is not the proof of local persistence. More generally, sufficiently small C^1 perturbations of the project-type distribution and network functions preserve the result conditional on positive cutoff densities, the same central allocation ordering, strict curvature, and local nonsingularity. Small asymmetric perturbations of the action state are covered by the same continuity logic.

D Additional Tables

The parameter table is generated from the same machine-readable result layer as Tables 1 and 2.

Table 3: Benchmark parameters

Parameter	Value
alpha	0.5
beta	0.05
gamma	0.9
k	0.27
kT	0.02
rho	0.15
rhoT	0.05
tau	0.05
v	0.1